

Manual

Monitoring of Shop Floor Data BDE-MAB 8.1

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1 Overview – Shop Floor Data / Order Data Monitoring

Overview

Purpose

The shop floor data / order data monitoring function provides numerous functions and overviews to monitor the orders / operations existing in the system. For this, the master data of the order / operation are accessed and compared with recorded data to provide an overview of the progress.

Implementation Notes

The shop floor data / order data monitoring function is used when

- You wish to obtain an overview of the orders / operations existing in the system.
- You wish to monitor the order progress for these orders / operations on the basis of the uploads recorded in the system.
- You wish to obtain an overview of the orders / operations completed in the previous shifts, the times required for this and the quantities produced.

Integration

The shop floor data / order data monitoring function uses order data that were either created directly in the system or that were transferred from other systems via interfaces.

Times and quantities entered in the BDE or MDE are accessed to display order progress.

Features

- Order overview:
 - Order overview as central overview function for the display of orders and their operations using a variety of selection criteria
 - Display of the order progress, i.e. all prepared, current, interrupted and completed OPs of a multi-level production order in the order overview
- Order information
 - Order information for the display of important information on an order, including its OPs, such as stock data, user fields, components, production resources and status information (degree of completion, remaining run time, quantities and times posted)
 - Display of comments entered at the terminal
- Operation overviews
 - Tabular list of the scheduled operations (status: prepared, interrupted)

- Tabular presentation of the currently logged on operations (status: running)
 - Tabular presentation of the operations already completed (status: completed)
 - Tabular presentation of all operations with predefined filter for display of logged on operations, completed operation and pool of orders
- Order shift log
 - Shift-related presentation of the completed operations
 - Graphic representation of quantities and times by article

2 Order Information

Overview

Menu	Production control → Production overview → Order information
Transaction code	orin
Function authorization	orin

Usage

Order information is the most important instrument for anyone in a position of responsibility in and around the production process, and it is used to monitor, control and oversee production.

Integration

Order information provides data, organized in a clear and well-structured manner, which are necessary when processing operations. This makes the order information one of the most important tools in the day-to-day production control and monitoring tasks. This application is especially indispensable for preparing work to be performed in the production department.

Selection criteria

The following selection criteria are available in the application:

Order

This selection criterion refers to the order number. The selected order is displayed with its operations.

Show split OPs

This selection criterion refers to the displayed operations. If this checkbox is enabled, the list of operations also includes split operations.

In addition, the below-mentioned information about the selected order is shown in the selection panel (view only):

Order status

Shows the status of the select order in text form (highlighted in color)

Archived

Provided that the order has already been archived, this is shown in text form behind the field displaying the order status.

Toolbar

In general, the parameters for calling the function or target application are taken over from the table. For this reason, an entry should always be selected before calling an application.

General index tab



Update order

The order can be updated by the [update order](#) function.



Schedule order

The order can be scheduled by the [schedule order](#) function.



Order overview

This button opens the application [order overview](#).



Documents

The documents defined at the operation can be displayed using this function. Documents are assigned in the "[edit production resources and tools](#)" function.



Serial numbers

Opens the application [edit serial numbers..\\functions\moc\MOC_OrderSerialNumber.pdf](#) for the order.

Edit index tab



Edit orders

Function authorization: edor.*

Calls up the application [edit orders](#) for the order.



Edit order sequences

Function authorization: edseq.*

Calls up the application [edit order sequences](#) for the order.



Edit operations

Function authorization: edop

Calls up the application [edit operations](#) for the order.



Edit operation long texts

Function authorization: edoptx.*

Calls up the application [edit operation long texts..\\.\functions\moc\MOC_EditOperationLongTexts.pdf](#) for the order.



Edit components

Function authorization: edopcomp.*

Calls up the application [edit components](#) for the order.



Edit production resources and tools

Function authorization: edopres.*

Calls up the application [edit production resources and tools](#) for the order.

Order data detail application

The master data associated with the orders is displayed in the order data detail application.

The separate fields in the order header are described [here](#)

The index tabs may deviate from the order used in the description.

Order data - status detail application

The order's current status is shown in the order data status detail application.

The times/ quantities represent the sum total of all operations.

Comments detail application

The BDE comments that were appended to the operations of the selected order during data entry are shown in this detail application. This supports efforts towards a paperless production process.

The following data is available for display:

Recording time

Point in time (date, time) when the comment was recorded.

Collected at workplace

Workplace at which the comment was recorded.

MES order number

The operation's combined order/ operation number that the comment was recorded for.

Person

Personnel number of the person according to the HR master data who recorded the comment.

Last name

Last name of the person according to the HR master data who recorded the comment.

First name

First name of the person according to the HR master data who recorded the comment.

Name

Entire name of the person (last name, middle name and first name) according to the HR master data who recorded the comment.

Operations detail application

The index tabs may deviate from the order used in this description.

Operations - operation detail application

The operation detail application provides a description of the operation's master data.

The fields for the operation are described [here](#).

Operation status - operation detail application

The current status for each operation is shown in the Status detail application.

The operation's status information is described [here](#).

Operation - RPA - operation detail application

The recorded times (actual values) for each operation broken down by RPA are shown in the detail application Operation - RPA.

Operations - logged on to workplace detail application

The detail application Logged on to workplace shows the workplaces that the operation selected at the top is logged onto. If an operation is logged on at several workplaces at the same time, the multiple workplaces are also shown here.

Workplace

Workplace

Workplace at which the operation is logged on.

Logon

Date

Point in time when the operation was logged on to the workplace.

Time

Point in time when the operation was logged on to the workplace.

Order

HYDRA order number

The selected operation's combined order/ operation number.

Operations - staff logged on detail application

Shown in this detail application are those employees that are logged on to the operation highlighted at the top. Because several employees can be logged on to one operation, in this case more than one person is shown for each operation here as well.

Person

Person

Personnel number of the person logged on at the operation according to the HR master data. The workplace at which this person with this operation is logged on can be seen in the workplace field.

Staff badge number

Staff badge number of the person logged on at the operation according to the HR master data.

Last name

Last name of the person logged on at the operation according to the HR master data.

First name

First name of the person logged on at the operation according to the HR master data.

Name

Entire name of the person (last name, middle name and first name) logged on at the operation according to the HR master data.

Logon

Date

Point in time when the operation was logged on to the workplace.

Time

Point in time when the operation was logged on to the workplace.

Advance logon

If the logon is currently still an advance logon recorded before the start of a shift, the flag in this column illustrates this.

Workplace

Workplace

Workplace at which the person with the currently selected operation is logged on to.

Operator position

Operator position (identification) that the person uses to log on to the workplace (only relevant if a person logs on at the Windows terminal with an operator position).

Order

HYDRA order number

The selected operation's combined order/ operation number.

Operations - resources logged on detail application

Shown in the detail table Resources logged on are the tools and other resources needed for production that are logged on either explicitly or implicitly at the operation.

Operations - material in use detail application

Shown in the Material in use detail application are the input material logged on at the operation selected on top.

Material

Batch number

HYDRA batch number of the material in use.

Material

Material number for the material in use according to the HYDRA batch inventory.

Material designation

Material designation for the material in use according to the HYDRA batch inventory.

Material type

Material type for the material in use according to the HYDRA batch inventory.

Quantity**Quantity**

Original quantity of the material in use according to the HYDRA batch inventory.

Remaining quantity

Remaining quantity of the material in use according to the HYDRA batch inventory.

Unit

Quantity unit of the material in use according to the HYDRA batch inventory.

Workplace**Workplace**

Workplace at which the input material/ batch is logged on.

Logon**Date**

Point in time when the input material/ batch was logged on to the workplace.

Time

Point in time when the input material/ batch was logged on to the workplace.

Order**HYDRA order number**

The selected operation's combined order/ operation number.

Operations - material produced detail application

Shown in this detail application is the output material (output batch) that is currently being produced during the operation.

Material**Batch number**

HYDRA batch number of the material produced.

Material

Material number for the material produced according to the HYDRA batch inventory.

Material designation

Material designation for the material produced according to the HYDRA batch inventory.

Material type

Material type of the material produced.

Quantity**Quantity**

Quantity recorded so far and posted to the material.

Unit

Quantity unit of the material produced.

Workplace**Workplace**

Workplace at which the output material/ batch is logged on.

Logon**Date**

Point in time when the output material/ batch was logged on to the workplace.

Time

Point in time when the output material/ batch was logged on to the workplace.

Order**HYDRA order number**

The selected operation's combined order/ operation number.

Operations - production resources and tools detail application

Shown in the production resources and tools detail application are all resources for an operation that were transferred via an interface. For operations that are not logged on, displayed here can also be those tools and resources not logged on to the operation.

Resource**Resource type**

Resource type for the resource, referred to as production resources and tools

Resource

Identification

Designation

Designation of the production resource according to the list of production resources and tools.

Comment 1, comment 2

Comment fields for the production resource according to the list of production resources and tools.

File**File name**

If the production resource is a file (image, document), the file name is shown here.

Path

Logical reference to the path configuration

Quantity**Required quantity**

Required quantity of the production resource or tool.

QU required quantity

Quantity unit of the required quantity of the production resource or tool.

Order**MES order number**

The selected operation's combined order/ operation number.

Operations - components detail application

Shown in the components detail application are all of the consumption materials for an operation that were transferred via an interface.

The fields for the components are described [here](#).

3 Order Overview

Summary

Menu	Production control --> Production overview --> Order overview
Transaction code	orov
Function authorization	orov

Utilization

The order overview function is of interest to all users who would like to get an overview of the production progress of an entire order, without having to do without progress details of individual operations.

Integration

The workplace overview is a function of production management that represents a production overview. The order overview is the central dialog to have information about HYDRA orders displayed. It shows dates, quantities and times of production orders, overhead cost orders, maintenance orders and all other order types. Orders are displayed in a table. The progress of an order is displayed by selecting a corresponding table row.

Selection criteria

If several selection criteria are used overlapping results are displayed in the order overview.

The application provides the following selection criteria:

Order

This selection criterion refers to the order number. The selected order is displayed. Wildcards may be entered.

Order type

This selection criterion refers to the order type at the order header. Only orders assigned to the selected order type(s) are displayed.

Category

This selection criterion refers to the category of the order type at the order header. Only orders assigned to the selected category(ies) are displayed.

Finished article

This selection criterion refers to the article in the order header. All orders assigned to the selected article are displayed. Wildcards may be used.

Article designation

This selection criterion refers to the article designation defined in the order header. All orders assigned to the selected article designation are displayed. Wildcards may be used.

Order status

This selection criterion refers to the order statuses at the order header. Only orders assigned to the selected order statuses are displayed.

Control

This selection criterion refers to the control indicator at the order header. Only orders assigned to the selected control indicators are displayed.

Sales order

This selection criterion refers to the sales order defined at the order header. All orders assigned to the selected sales order are displayed. Wildcards may be used.

Customer designation

This selection criterion refers to the customer designation defined in the order header. All orders including the selected customer designation are displayed. Wildcards may be used.

Project number

This selection criterion refers to the project number defined in the order header. All orders that are assigned to the selected project number are displayed. Wildcards may be used.

Planned order

This selection criterion refers to the planned order defined in the order header. Wildcards may be used.

Cost object

This selection criterion refers to the cost object defined in the order header. All orders of the selected cost object are displayed. Wildcards may be used.

MRP controller

This selection criterion refers to the MRP controller defined in the order header. All orders of the selected MRP controller are displayed.

Order group

This selection criterion refers to the order group defined in the order header. All orders that are assigned to the selected order group are displayed.

Basic start date ... Basic start date

This selection criterion refers to the basic start date defined in the order header. Only are orders displayed, which are planned on or between the selected basic start dates.

Basic end date ... Basic end date

This selection criterion refers to the basic end date defined in the order header. Only are orders displayed, which are planned on or between the selected basic end dates.

Job end ... Job end

This selection criterion refers to the actual job end of the order header. Orders, which are dated on or between the selected order end dates, are displayed only.

Priority ... Priority

This selection criterion refers to the priority defined in the order header. All orders having the selected priority are displayed.

Order index ... Order index

This selection criterion refers to the order index defined in the order header. All orders having the selected order index are displayed.



The responsibility area is not checked in this application.

"Order overview" detail application**"Status" category****Order status**

The bitmap ("LED") defined in the status configuration is displayed as status.

In the HYDRA standard the color of the status LED corresponds to the one of the control LED.

Order status (text)

The status text results from the current status of the operation.

Status since

Date/time since when the order is in this status.

Status since time

Date when the status was set

Order start

Date/time of starting the order, i.e. date when the first operation of an order starts.

Order end

Date/time when the order is finished, i.e. date when the last operation of an order is logged off.

"Order" category

Shows specific data for [orders](#).

Relevant fields are:

Order type

Displays the order type as text and symbol. The order types within the HYDRA standard are described in the glossary. Further order types can be defined while HYDRA is customized.

Order

Number of the corresponding order

User fields of the order header

Shows the configured user fields for the order overview in the "user fields" category.

"Progress" category**Number of OPs**

Number of operations of the corresponding order.

Finished OPs

Number of finished operations of the corresponding order.

"Quantities" category**Target quantity (B)**

Quantity specification for the order in base quantity unit.

Target scrap

Target scrap of the order in base quantity unit.

Unit

Defined unit (base quantity unit)

Yield (B)

Recorded yield in base quantity unit of the last operation that can be recorded.



This is the last operation included in the order network that is neither locked (internal control flag "Y") nor deleted logically (internal control flag "D"). This operation "provides" the quantity for the entire order.

This means that yield is 0 as long as the order has not been finished, i.e. as long as no quantity has been posted onto the last operation that can be recorded for the order.



Provided that the last operation that can be recorded has a quantity > 0, but this quantity is not shown in the order overview, please check the procedure described in the document entitled [Activating_OrderRelatedKeyfigures.pdf](#) or proceed as described there.

Scrap (B)

Sum total of the scrap quantities entered for all operations posted of the order in base quantity unit.

Prerequisite: scrap has been posted in base quantity unit to the operations.

Rework (B)

Sum total of the rework quantities entered for all operations posted of the order in base quantity unit. Prerequisite: the rework quantity has been posted in base quantity unit to the operations.

Open quantity (B)

Sum total of the open quantities entered for all operations posted of the order in base quantity unit. Prerequisite: the open quantity has been posted in base quantity unit to the operations.

"Target times" category**Planned lead time**

The planned lead time includes all planned execution times, like setup time, processing time, inspection time and dismantling/teardown time

Target setup time

Target setup time for the operation. The setup time is transferred by the ERP system or can be calculated using a formula that is defined while HYDRA is customized.

Planned processing time

Target processing time for the operation. The processing time is transferred by the ERP system or can be calculated using a formula defined while HYDRA is customized.

Planned execution time

Total of planned setup time + planned processing time

Labor utilization

Total of the planned labor utilization of all enterable, active OPs

"Actual times" category**Retention period of order**

The retention period of the order results from the period of time between the first transfer of the order from the PPS system ("order release" = creation date of the order header in HYDRA) and the actual logoff of the last active operation of the order (from a chronological view).

Please note:

- It cannot be recognized or considered whether this point in time of the order transfer is the point in time when the order is transferred for the first time from the PPS system or whether the order has meanwhile been deleted and resent due to technical problems.
- In case an order is transferred several times and the previous order is deleted with each transfer, the creation date of the order header is that of the last transfer (from a chronological view).

Lead time

The order duration results from the period of time between the first logon of an operation of the order and logoff of the last operation that is active (from a chronological view).

Actual setup time

The setup time of the order is the total of setup times (RPA 7) of all enterable, active operations.

Actual processing time

The processing time of the order is the total of main utilization times (RPA 11) of all enterable, active operations.

Actual downtimes

The downtime of the order is the total of downtime times (RPA 1 to 6, RPA 8 to 10) of all enterable, active operations.

Assignment time

The assignment time is the total of the setup time (RPA 7), processing time (RPA 11) and downtimes (RPA 1...6, RPA 8...10) of all enterable, active operations.

Labor utilization

The labor utilization of the order is the total of labor utilization times of all enterable, active operations.

"Actual dates" category

Order start date

Date when the order was started, i.e. when the first operation of the order was logged on.

Order start time

Time when the order was started, i.e. when the first operation of the order was logged on.

Order end date

Date when the order was finished, i.e. when the last operation of the order was logged off.

Order end time

Time when the order was finished, i.e. when the last operation of the order was logged off.

"Key figures" category

Rate of capacity utilization

The capacity utilization rate is the relation between the processing time (RPA 11) and the assignment time (RPA 1 ... 11) in percent.

$$NGRAD = 100.0 * \frac{ANR.EGR : BMK11}{\sum_{n=1}^{11} ANR.EGR : BMKn}$$

Setup rate

The setup rate is the relation between the setup time/costs (RPA 7) and the assignment time (RPA 1...11) in percent.

"Order progress" detail application

The "order progress" detail application shows all operations pertaining to the above selected order at a glance.

Provided that several orders are selected in the order overview, all operations of these selected orders are displayed here. In this case, it is recommendable to show the "order" column.

"Status" category

Status

The bitmap ("LED") defined in the status configuration is displayed as operation status.

In the HYDRA standard the color of the status LED corresponds to the one of the control LED.

Status text

The status text results from the current status of the operation.

Status since

Time since that the current status has been pending.

The field is empty for prepared operations.

Predecessor status

Status of the predecessor operation. Here, it can be recognized whether the predecessor operation has already been started and thus material, which will be further processed in the current operation, has already been processed or produced.

Secondary status

Displays the currently set secondary status.

Please note: Secondary statuses are optionally configured and used while HYDRA is customized.

"Order" category

Order

Number of the corresponding order

"Operation" category

Shows specific data for [operations](#).

Note

Short text of the first note of an operation.

"Primary quantity/secondary quantity/tertiary quantity" category

Target quantity

Quantity specification for the [operation](#).

Yield

The yield entered at the terminal or console/MOC is displayed in the yield column.

Scrap

The scrap entered at the terminal or console/MOC is displayed in the "scrap" column.

Rework

Quantity that has to be reworked.

Problem quantity

The problem quantity represents another quantity account.

Unit

Quantity unit of the displayed values.



The quantities listed here are displayed as base, primary, secondary and tertiary quantity. In the majority of cases, it is reasonable to have only one of these quantity types displayed. The primary quantity is the quantity type in which values are entered at the terminal.

"Postings" category**First logon**

Date/time when the operation is logged on first

Last interruption

Date/time when the operation was interrupted at last

Last logoff

Date/time when the operation was logged off at last

Last logon

Date/time when the operation is logged off at last

"Specifications for production" category**Tool, DNC, material**

The component list shows the components that are assigned to the operation

Relevant fields are:

Planned workplace

Workplace on which the operation is planned.

Planned for

The "planned" column shows whether the operation is planned

G on a group (pool of groups) or

M on a workplace (pool of workplaces/machines).

Group

Group which the operation is planned for.

"Dates" category

Date specifications for the [operation](#)

"Miscellaneous" category**Upload number**

Upload number of the [operation](#)

"Target times" category

Default target time specifications for the [operation](#)

Additionally calculated fields:**Total setup time**

Total of target setup time, additional setup time and target dismantling/teardown time.

Planned execution time

Sum total of the target setup time (total) and target processing time.

"Actual times" category**Setup time**

Posted setup time of the operation.

Execution time

Sum total of the total setup time and processing time.

Processing time

Posted processing time (RPA 11) of the operation.

Downtime

Posted downtime of the operation

Total of downtimes (RPA 1..6, RPA 8..10)

Assignment time

The assignment time is the total of the setup time (RPA 7), processing time (RPA 11) and downtimes (RPA 1...6, RPA 8...10).

Lead time

Posted assignment time of the operation. The lead time results from the period of time between the first logon of an operation and the last logoff of the operation.

Remaining run time

Still remaining production time. This is a calculatory value that is calculated by a formula based on different parameters. The formula is defined in the operation. (RRT 1)

Remaining run time 2

Still remaining production time. This is a calculatory value that is calculated by a formula based on different parameters. The formula is defined in the operation. (RRT 2)

Processing category

In addition to further target specifications for the [operation](#), the actual cycle relating to operations is also provided here.

Actual cycle

The actual cycle is calculated and displayed as follows:

- 1) OP prepared: By definition, the actual cycle is = 0
- 2) Provided that the OP is logged on, the actual cycle is the current actual cycle from machine data collection (depends on the machine connection). This one is also displayed in the application [workplaces/machines](#).
- 3) An average cycle time is calculated when an operation is interrupted or logged off/finished. This average cycle time results from the division of the previously posted production time (RPA 11) by the previously recorded strokes . Only these order postings result in a calculation (this is not the case, for example, after changes have been made in the maintenance of postings).

User fields of the operation/order header

It is possible to display the user fields of operations and order headers in tables. It is necessary to use a certain user field configuration for the structure of the columns as orders with different user field configurations can be displayed simultaneously in the order overview:

Operation: Object AGNR, user field key SYSTEM

Order header: Object AUNR, user field key SYSTEM

A category is generated for user fields in the table. All defined columns are initially generated in the relevant category. The position and whether or not these columns should be visible can be edited using the column configurator. The entries are configured while HYDRA is customized.

Toolbar

The function authorization required to execute the relevant function is entered in parentheses.

"Order" tab



Order status (function authorization: or.statchg)

Function to [change the order status](#)



Update order (function authorization: or.actualize)

The "[update order](#)" function allows for one or several selected orders to be updated.



Schedule order (function authorization: or.terminate)

The "[schedule order](#)" function allows for one or several selected orders to be scheduled.



Order information (function authorization: or.in)

Calling up the [order information](#)

"Operation" tab



Order status (function authorization: op.statchg)

Function to [change the operation status](#)



Secondary status (function authorization: op.secstatchg)

Function to [change the secondary status of an operation](#)



Change resource status (function authorization: op.resstatchg)

Function to [change the resource status of an operation](#)



Lock operation (function authorization: op.lock)

The [lock operation](#) button allows for one or several selected operations to be blocked



Unlock operation (function authorization: op.unlock)

The [unlock operation](#) button allows for one or several selected operations to be unlocked

Please note

If the HYDRA module MPL (material and production logistic) is used batch-related order postings cannot be carried out at the HYDRA console/MOC due to their complexity.



Log on (function authorization: op.logon)

Operations can be logged on to the system using the "[log on](#)" function

**Partial confirmation/upload (function authorization: op.partconf)**

The "[partial upload](#)" function allows for partial uploads on operations to be recorded in the system

**Interrupt (function authorization: op.interrupt)**

Operations can be interrupted in the system using the "[interrupt](#)" function

**Log off (function authorization: op.logoff)**

Operations can be logged off from the system using the "[log off](#)" function

**Terminate (function authorization: op.finish)**

Interrupted or prepared operations can be logged off from the system using the "[terminate](#)" function

**Reactivate (function authorization: op.reactivate)**

Terminated operations can be reactivated in the system using the "[reactivate](#)" function

"Person" tab

Please note

The functions relating to operations "log on/interrupt/log off" are to be used for posting data to group workplaces.

**Log person on (function authorization: pn.logon)**

A person may be logged on to an operation/machine using the [log person on](#) function

**Log person off (function authorization: pn.logoff)**

A person may be logged off from the corresponding operation/machine using the [log person off](#) function

4 Order Shift Log

Summary

Menu	Order Management --> Production Reports --> Order Shift Log
Transaction code	ospr
Function authorization	ospr

Utilization

The order shift log is a must for every shift supervisor and foreman. It provides information that people who are in charge of production need for their daily reports.

The order shift log is a function for production management. The individually configurable user interface presents shift-specific order data in a clear and comprehensive way.

The shift log analyses the collected order data (quantities and times) in relation to the entered shifts.

Integration

The order shift log shows all operations that meet the restrictions made in the selection panel. In this context, only BDE log records of the record type "U" and "E" are taken into account.

This function provides the operations that are produced within one shift. In addition to the produced quantities, the times that have accrued accordingly are displayed as well. This applies to each order / operation.

The postings relating to HYDRA-BDE function allows for the recorded order data to be corrected.

Selection criteria

The application provides the following selection criteria. When data is requested, the responsibility area of the workplace which the BDE log records have been posted onto is checked.

Date from ... to ...

The entered period of time restricts the selection of log records. Log records the start date of which (logon date) coincides with the specified period of time are selected.

Default setting from "yesterday" until "today". The calculation is based on the Gregorian calendar.

Shift: all, 1, 2, 3, 4

Within the entered period of time only those log records are selected that are assigned to the entered shift according to the shift model.



Operations logged on to the current shift at the time of data selection are not taken into consideration, as log records have not yet been generated for them.

Workplace from... to ...

Those log records are selected that are posted onto the entered workplace and which the user is authorized for via the responsibility area of the workplace. It is possible to use wildcards in the "from" field.

Group from ... to ...

Those log records are selected that are posted on workplaces which the user is authorized for via the responsibility area of the workplace and that are assigned to the specified group. It is possible to use wildcards in the "from" field.

Cost center

Those log records are selected that are posted onto workplaces that the user is authorized for via the responsibility area of the workplace and that are assigned to the specified cost center. Wildcards may be used.

Report group

Report group

Order

Those log records are selected that are posted onto the specified order.

Operation

Those log records are selected that are posted onto the specified operation.

Article

Those log records are selected that are posted onto operations with the specified article.

Tool

Those log records are selected that are posted onto operations with the specified tool.

"Order shift log" detail application

The table shows the following fields

"Shift" category**Shift date**

Shift date of the shift in which the operation was produced. (Basis: BDE log record)

Shift

Shift in which the operation was produced. (Basis: BDE log record)

"Order" category**Category**

Order category of the order/operation, e.g. production order (FA) or overhead cost order (GK).
(Basis: pool of orders).

Order type

Order type of the order. (Basis: pool of orders)

Order

Order number of the order/operation. (Basis: pool of orders)

Sequence

Sequence number of the operation. (Basis: pool of orders, subject to customizing/configuration)

Operation

Operation number of the operation. (Basis: pool of orders)

Split

Split number of the operation, provided that it is a split operation. (Basis: pool of orders, subject to customizing/configuration).

Operation designation

Designation of the operation. (Basis: pool of orders)

Article, article designation

Article number and article designation of the operation. (Basis: pool of orders)

"Workplace" category**Workplace, group, cost center**

In addition to the order number and article number, the workplace (including group and cost center of the workplace) is displayed on which this operation was produced within the selected shift.
(Basis: workplace: BDE log record; group, cost center: workplace/resource master data)

"Primary quantity" category**Target quantity**

Total target quantity of the operation in the relevant quantity unit (primary quantity unit, secondary quantity unit, tertiary quantity unit, base quantity unit; basis: pool of orders).



This column is not added up. Since totals formation is not correct if one and the same operation is produced during several shifts.

Target quantity per shift

This field includes the theoretical target quantity for the posting. The duration of the posting may even be shorter than the shift duration if the operation, for example, was not logged on over the entire shift. If the OP is logged on over the entire shift, the target quantity will be calculated for each shift; otherwise the target quantity will only be calculated for the shorter posting period.

$$SSKGUT = \frac{\sum_{01}^{11} ANR.EGR:BMK}{\left(\frac{ANR.SZY}{1000.0} \right)} * ANRTL$$

Yield

The yield posted onto this operation – relating to the selected shift. (Basis: BDE log record).

Scrap

The scrap posted onto this operation – relating to the selected shift. (Basis: BDE log record).

Rework

The rework quantity posted onto this operation – relating to the selected shift. (Basis: BDE log record)

Open quantity

Open quantity posted onto this operation – relating to the selected shift. (Basis: BDE log record)

Quantity unit

Corresponding quantity unit. (Basis: BDE log record)

"Duration" category**Target duration**

The total target duration based on operations is determined as follows: (Basis: pool of orders)

((Target cycle of the operation [per 1000]) / 1000 / the operation's partitioning * target quantity of the operation in primary quantity unit) + setup time of the operation



This column is not added up. Since totals formation is not correct if one and the same operation is produced during several shifts.

Production

The production time recorded on this operation - relating to the selected shift. (Basis: BDE log record)

Downtime

The downtime recorded on this operation - relating to the selected shift. (Basis: BDE log record)

Sum

Total of the accrued production time and downtime (total of the columns "production + downtime")

"RPA" category

RPA

Detailed presentation of times recorded on the level of resource performance accounts . (Basis: BDE log record) .

"Key figures" category

Rate of capacity utilization

$$NGRAD = 100.0 * \frac{ANR.EGR : BMK11}{\sum_{01}^{11} ANR.EGR : BMK}$$

Output rate

$$AUSBGD = 100.0 * \left(\frac{ANR.EGR : GUTP}{1000.0 * \sum_{01}^{11} ANR.EGR : BMK * ANR.TLG} \right) * ANR.SZY$$

Scrap rate

$$AQUOTE = 100.0 * \frac{ANR.EGR : AUSEP}{(ANR.EGR : GUTP + ANR.EGR : AUSEP)}$$

Assignment utilization rate

$$BNGRAD = 100.0 * \frac{ANR.EGR : BMK11}{\left(ANR.EGR : BMK01 + ANR.EGR : BMK02 + ANR.EGR : BMK03 + ANR.EGR : BMK04 + ANR.EGR : BMK05 + ANR.EGR : BMK08 + ANR.EGR : BMK09 + ANR.EGR : BMK10 + ANR.EGR : BMK11 \right)}$$

Machine category

Target cycle

Time need that is planned for 1000 production cycles of the machine. This value is the default value for HYDRA-MDE machine monitoring (cycle monitoring).

Actual cycle

$$IZYCLC = 1000.0 * \frac{ANR.EGR : BMK11}{ANR.EGR : HUBG}$$

Partitioning

Integer value that defines the number of parts which are produced with a clocked pulse.

Total cycles

Total cycles

Please note

Display of actual quantities and actual durations (RPA)

No shift automatic takes place for workplaces that are either not assigned to a terminal or that are assigned to a terminal configured as terminal of the shop floor data collection module (HYDRA-BDE terminal). This means that when the shift ends, orders or staff are not posted automatically. As in this case, quantities and times cannot exactly be assigned to a shift, they are assigned proportionally in the shift log. This assignment is based on the shift calendar of the workplace.

Example:

Given is a shift model with shift 1: 6.00 am – 2.00 pm and shift 2: 2.00 pm – 10.00 pm. This shift model is assigned to a workplace matching the above-mentioned criteria. Moreover, an operation is logged on at 1.00 pm and an operation is logged off at 4.00 pm. 90 is confirmed/uploaded as yield when the operation is logged off.

Consequently, the shift log determines an order time of 60 minutes and yield “30” for the operation of shift 1. Shift 2 has an order time of 120 minutes and yield “60”. The times relating to RPA are also determined based on the shift model of the workplace.



Operations logged on to the current shift at the time of data selection are not taken into consideration, as log records have not yet been generated for them.

"Durations by article" detail application

The detail application “bar chart – durations” shows the posted durations accumulated on the article level. In this context, the operations selected in the detail application “order shift log” are taken into consideration.

The bar chart shows the article numbers of the selected operations on the y-axis and absolute values (durations) are displayed on the x-axis. The bar color is based on the respective quantity account (production / RPA 11: green, downtimes / RPA 1-11: red). Sorting is effected in descending order by production duration.

Using a multi combo box, it can be defined for which durations the bars are to be displayed:

- Production
- Downtimes

They are displayed in a “stacked” manner in order for the overall quantity to be defined according to the user’s requirements.

By checking the “show labels” option, the values are displayed on the bars. In this context, it has to be taken into account that these labels are displayed for each selected duration.

"Quantities by article" detail application

The detail application "quantities by article" shows the posted quantities (accumulated on the article level). In this context, the operations selected in the detail application "order shift log" are taken into consideration.

The bar chart shows the article numbers of the selected operations on the y-axis and absolute values (quantities) are displayed on the x-axis. Sorting is effected in descending order by production duration.

Toolbar

In general, the parameters for calling the function or target application are taken over from the table. For this reason, an entry should always be selected before calling an application.



Order information

This button opens the application [order information](#).



Order overview

This button opens the application [order overview](#).

5 Pool of Orders

Summary

Menu	Production control --> Planning aid --> Pool of orders
Transaction code	plop
Function authorization	plop

Utilization

The pool of orders application is the most important dialog for the shift foreman, supervisor and foreman.

The pool of orders function is a tool that provides users with a selection of operations, categorized clearly, each of which can be found in the pool for a machine or group.

Integration

All orders that match the selections entered are displayed in the pool of orders.

Generally, only unplanned and planned operations are displayed in the pool of orders. From system's standpoint, in this case they are operations that are marked with the control indicators V, U or S. Typically, they are operations with an operation status "prepared", "interrupted" or "not free" (depending on the project).

The following detail applications are an integral part of the pool of orders:

- Pool of orders
- Running operations
- Finished operations

Selection criteria

The following selection criteria are available in the application:

Workplace from... to ...

This selection criterion refers to the workplace defined in the operation. All operations are displayed that are assigned to the selected workplaces. You can also run a search using wildcards.

Group from ... to ...

This selection criterion refers to the group defined in the operation. All operations are displayed that are assigned to the selected group, irrespective of whether they are already planned in detail for a workplace or not. You can also run a search using wildcards.

Planned for

You can use this option to restrict whether you would like to display operations that have already been planned in detail for a workplace or only operations that are still in the pool of groups.

The column "planned for" displays whether the operation is planned

- G for a group (pool of groups), or
- M for a workplace (workplace/ machine pool)

Order

This selection criterion refers to the order number. Operations are displayed that contain the order number entered. You can also run a search using wildcards.

Article

This selection criterion refers to the article in the operation. All operations are displayed, to which the article entered has been assigned. You can also run a search using wildcards.

Article designation

This selection criterion refers to the article in the operation. All operations are displayed that match the article designation entered. You can also run a search using wildcards

Operation status

Current operation status

Predecessor status

Status of the preceding operation.

Control

The current production indicator for the operation

Category

This selection criterion refers to the order type category of the operation. Only those operations are displayed with an order type that matches the category entered.

Order type

This selection criterion refers to the order type of the operation. Only operations are displayed that have the order type selected.

Processing code

This selection criterion refers to the processing code at the operation. Only operations are displayed that contain the processing code selected. You can also run a search using wildcards.

Priority

This selection criterion refers to the priority of the operation. Only operations are displayed that have the priority selected.

OP

This selection criterion refers to the operation number. Operations are displayed that have the operation number entered. You can also run a search using wildcards

Tool

This selection criterion refers to the tool used in the operation. All operations are displayed that have been assigned with the tool entered. You can also run a search using wildcards.

Show split OPs

If this box is checked, split operations are also displayed.

Customer name

This selection criterion refers to the customer designation defined in the operation. All orders are displayed that contain the customer designation selected.

Cost center

This selection criterion refers to the cost center of the workplace defined in the operation. All operations are displayed that are assigned to the selected cost center. You can also run a search using wildcards.

Planned start between ... and

This selection criterion refers to the planned start date defined in the operation. Only operations are displayed that are planned on or between the selected planned start date. Whether they are planned in detail to a workplace or are still in the pool of groups is irrelevant.

Scheduled start between ... and

This selection criterion refers to the scheduled start date of the operation. Only operations are displayed that are planned on or between the selected scheduled start date.

Earliest start between ... and

This selection criterion refers to the earliest start date of the operation. Only operations are displayed that are planned on or between the selected earliest start date.

Latest end between ... and

This selection criterion refers to the latest end date of the operation. Only operations are displayed that are planned on or between the selected latest end date.

Actual start between ... and

This selection criterion refers to the start date of the order. Only operations are displayed with an order start that is on or between the selected date.

Actual end between ... and

This selection criterion refers to the finish date of the order. Only operations are displayed with an order end that is on or between the selected date.

Basic start date between ... and

This selection criterion refers to the basic start date of the order. Only operations are displayed with a basic start date for an order that is on or between the selected date.

Basic end date between ... and

This selection criterion refers to the basic end date of the order. Only operations are displayed with a basic end date for an order that is on or between the selected date.

Sales order

This selection criterion refers to the sales order defined in the order header. All operations are displayed that contain the selected sales order.

Order index from ... to

This selection criterion refers to the order index in the order header. All operations are displayed that have an order index defined in the order header that matches the selected order index.

Planned order

This selection criterion refers to the planned order number from SAP that is defined in the order header.

Project number

This selection criterion refers to the project number in the operation. All operations are displayed that are assigned to the selected project number.

Order group

This selection criterion refers to the order group in the order header. All operations are displayed with an order group in the order header that is assigned to the selected order group. You can also run a search using wildcards.

MRP controller

This selection criterion refers to the MRP controller defined in the order header. All operations are displayed with an MRP controller in the order header that matches the MRP controller entered. You can also run a search using wildcards.

MOP

Restricted to a merged operation.
You can also run a search using wildcards.

Show MOP

Definition of the [merged operations](#) to be displayed.

Initially, merged operations and individual operations are displayed; operations combined into an MOP are not displayed.



In this application, the responsibility area of the planned operations is checked against the workplace, which they are planned for.

"Pool of orders" detail application

The following fields are displayed in the table

"Status" category**Status**

The bitmap ("LED") defined in the status configuration is displayed under status.

In this column, the number of operations is displayed in the total line.

Status text

The status text is based on the current status of the operation.

Status since

Information shown as of when the corresponding status was set.

Predecessor status

Status of the preceding operation. It shows whether the preceding operation was already started and if as a result, material was already processed or produced that is now being further processed in the current operation.

Secondary status

The currently set secondary status is displayed here.



The ability to configure and use the secondary status is offered as another option for customized systems.

"Order" category

Specific data is displayed here for [operations](#) and [orders](#).

Relevant fields are:

Order type

Order type displayed as text and icon.

The order types available in HYDRA standard are described in the glossary. Additional order types can be defined during the HYDRA customizing process.

Order

The number for each order.

Sequence

Order sequence (only relevant when using sequences).

OP

The number for each operation.

Split

Split number, if the operation is split.

Current number of splits

Number of splits in an operation, if the operation was split.

Note

Short text of the first note of an operation.

Printed

This column shows if the corresponding time ticket has already been printed.

"Specifications for production" category

Fields relevant to this are:

Planned workplace

Workplace at which the operation is planned.

Planned for

Displayed in the column "planned" is whether the operation is planned as

G at a group (pool of groups), or

M at a workplace (pool of workplaces/ machines).

Group

Group for which the operation is planned.

"OP dates" category

Date specifications for the [operation](#)

"Remaining run time" category**Remaining run time**

Production time that still remains. This is a calculated value that is determined using a formula based on various parameters. The formula is defined at the operation.

"Target times" category

Default target time specifications for the [operation](#)

Additionally calculated fields:**Total setup time**

The sum of setup time, additional setup time and dismantling time.

Target execution time

The sum total made up of total setup time and processing time.

"Primary quantity/secondary quantity/tertiary quantity" category**Target quantity**

Quantity defined for the [operation](#)

Yield

Yield posted onto the operation.

Scrap

Scrap posted onto the operation.

Rework

The quantity that still has to be reworked and that is posted onto the operation.

Problem quantity

The problem quantity posted onto the operation.

Unit

Quantity unit for the values displayed.



The quantities listed here are displayed as basic, primary, secondary and tertiary quantities. In most cases, you can only display one of these quantity types. The quantity type used to enter quantities at the terminal is the primary quantity.

Toolbar**Shop floor papers**

[Print shop floor papers](#) function

**Time ticket**

[Print time tickets](#) function

**Generate merged operation**

[Generate merged operation](#) function

Selected operations are consolidated to a single merged operation. The consolidated operation is representatively logged on and displayed at the terminal.

**Cancel merged operation**

[Cancel merged operation](#)

The merged operation is broken up into separate operations.

**Order information**

This button opens the application [order information](#).

**Order overview**

This button opens the application [order overview](#).

**Schedule controlling: save baseline plan**

[Schedule controlling: save baseline plan](#) function

6 Operations Logged On

Summary

Menu	Production control --> Production overview --> Running operations (operations logged on)
Transaction code	rop
Function authorization	rop

Utilization

The "running operations" function shows the currently running and automatically interrupted (by change of shift) operations (with the control indicators L, F).



Please note in this context that registered operations (status "running") are only shown by their workplace to which they are logged on if the option "Show the operations logged on for each workplace" is set.

Selection criteria

The application provides the following selection criteria:

Workplace from... to ...

This selection criterion refers to the workplace planned for the operation. All operations assigned to the selected workplaces are displayed. It is possible to use wildcards.



For operations that are logged on, this is not the workplace which the operation is currently logged on to but the workplaces which the operation is planned for.

Group from ... to ...

This selection criterion refers to the group that is planned for the operation. All operations assigned to the selected group are displayed. It is possible to use wildcards.



For operations that are logged on, this is not the group of the workplace which the operation is currently logged on to but the group which the operation is planned for.

Planned for

This option allows for the displayed operations to be restricted to operations that have initially been planned for a workplace or a group.

Order

This selection criterion refers to the order number. Operations assigned to the specified order number are displayed. It is possible to use wildcards.

Article

This selection criterion refers to the article in the operation. All operations assigned to the specified article are displayed. It is possible to use wildcards.

Article designation

This selection criterion refers to the article in the operation. All operations assigned to the specified article designation are displayed. Wildcards can be used.

Operation status

Current operation status

Predecessor status

Status of the predecessor operation.

Control

Current production flag of the operation

Category

This selection criterion refers to the category of the operation's order type. Only operations are displayed the order types of which match the specified category.

Order type

This selection criterion refers to the order type of the operation. Only operations assigned to the selected order type are displayed.

Processing code

This selection criterion refers to the processing code of the operation. Only operations assigned to the selected processing code are displayed. It is possible to use wildcards.

Priority

This selection criterion refers to the priority of the operation. Only operations assigned to the selected priority are displayed.

OP

This selection criterion refers to the operation number. Operations assigned to the entered operation number are displayed. It is possible to use wildcards.

Tool

This selection criterion refers to the tool in the operation. All operations assigned to the specified tool are displayed. It is possible to use wildcards.

Show split OPs

If this checkbox is checked split operations are displayed additionally.

Customer designation

This selection criterion refers to the customer designation defined in the operation. All orders including the selected customer designation are displayed.

Cost center

This selection criterion refers to the cost center of the workplace defined in the operation. All operations assigned to the selected cost center are displayed. It is possible to use wildcards.

Planned start between ... and

This selection criterion refers to the planned start date defined in the operation. Only operations which are planned on or between the selected planned start dates are displayed.

Scheduled start between ... and

This selection criterion refers to the scheduled start date of the operation. Only operations planned on or between the selected scheduled start dates are displayed.

Earliest start between ... and

This selection criterion refers to the earliest start date of the operation. Only are operations displayed, which are planned on or between the selected earliest start dates.

Latest end between ... and

This selection criterion refers to the latest end date of the operation. Only operations planned on or between the selected, latest end dates are displayed.

Actual start between ... and

This selection criterion refers to the start date of the order. Only operations are displayed the order start of which is planned on or between the selected dates.

Actual end between ... and

This selection criterion refers to the end date of the order. Only operations are displayed the order end of which is planned on or between the selected dates.

Basic start date between ... and

This selection criterion refers to the basic start date of the order. Only operations are displayed whose basic start time of the order is planned on or between the selected dates.

Basic end date between ... and

This selection criterion refers to the basic end date of the order. Only operations are displayed whose basic end time of the order is planned on or between the selected dates.

Sales order

This selection criterion refers to the sales order defined at the order header. All operations assigned to the selected sales order are displayed.

Order index from ... to

This selection criterion refers to the order index defined at the order header. All operations are displayed whose order index at the order header matches the selected order index.

Planned order

This selection criterion refers to the planned order number from SAP that is defined in the order header.

Project number

This selection criterion refers to the project number in the operation. All orders that are assigned to the selected project number are displayed.

Order group

This selection criterion refers to the order group at the order header. All operations are displayed whose order group at the order header is assigned to the selected order group. It is possible to use wildcards.

MRP controller

This selection criterion refers to the MRP controller defined for the order header. All operations are displayed whose MRP controller of the order header corresponds to the entered MRP controller. It is possible to use wildcards.

MOP

Restricts the merged operations that are displayed. Wildcards can be used.

MOP display

Defines the [merged operations](#) to be displayed. At first, merged operations and individual operations are displayed. Operations summarized in an MOP are not displayed.

Show the operations logged on for each workplace

If this option is checked, an operation that is currently being logged on to multiple workplaces will be shown as many times as it is logged on.

The category "logged on to workplace" (hidden by default) shows information about the workplace which the operation is logged on to.



In this application, the responsibility area of the planned operations is checked against the workplace which they are planned for.

"Running operations" detail application

The table shows the below-mentioned fields, amongst other things. Information on further fields can be found in the documents dealing with

- [data about the backlog of orders](#)
- [current data and key figures on the operation](#)

"Status" category

Status

The bitmap ("LED") defined in the status configuration is displayed as status.

The number of operations is displayed in this column within the totals line.

Status text

The status text results from the current status of the operation.

Status since

Informs about the time since when the corresponding status is set

Predecessor status

Status of the predecessor operation. Here, it can be recognized whether the predecessor operation has already been started and thus material, which will be further processed in the current operation, has already been processed or produced.

Secondary status

Displays the currently set secondary status.



Secondary statuses are configured and used optionally while customizing HYDRA.

"Order" category

Displays specific data for [operations](#) and [orders](#).

Relevant fields are:

Order type

Displays the order type as text and symbol. The order types of the HYDRA standard are described in the glossary. Further order types can be defined while customizing HYDRA.

Category

The category summarizes similar order types. The following categories are available in HYDRA:

FA = production order

PJ = project order

PM = maintenance order

KP = capacity order

GK = overhead cost order

Order

Number of the corresponding order

Sequence

Order sequence (only relevant if sequences are used)

OP

Number of the respective operation

Split

Split number provided that it is a split operation.

Current number of splits

Number of splits of an operation, provided that this one has been split.

Note

Short text of the first note of an operation.

Printed

This column shows whether the corresponding time ticket has already been printed.

"Specifications for production" category

Relevant fields are:

Planned workplace

Workplace on which the operation is planned.

Planned for

The "planned" column shows whether the operation is planned

G on a group (pool of groups) or

M on a workplace (pool of workplaces/ machines).

Group

Group which the operation is planned for.

"OP dates" category

Date specifications for the [operation](#)

"Remaining run time" category**Remaining run time**

Still remaining production time. This is a calculatory value that is calculated by a formula on the basis of different parameters. The formula is defined in the operation.

"Target times" category

Default target time specifications for the [operation](#)

Additionally calculated fields:**Total setup time**

Total of setup time, additional setup time and dismantling/teardown time.

Planned execution time

Sum total of the total setup time and processing time.

"Primary quantity/secondary quantity/tertiary quantity" category**Target quantity**

Quantity specification for the [operation](#).

Yield

The yield entered at the terminal or console/MOC is displayed in the yield column.

Scrap

The scrap entered at the terminal or console/MOC is displayed in the "scrap" column.

Rework

Quantity that has to be reworked.

Problem quantity

The problem quantity is another quantity account.

Unit

Quantity unit of the displayed values.



The quantities listed here are displayed as base, primary, secondary and tertiary quantity. In the majority of cases, it is reasonable to have only one of these quantity types displayed. The primary quantity is the quantity type in which values are entered at the terminal.

"Postings" category**First logon**

Date and time when the operation was logged on for the first time.

Last logoff

Date and time when the operation was logged off the last time. This kind of information can be interesting if an operation was reactivated.

Date of last posting

Date of the last posting made for this operation.

Time of last posting

Time of the last posting made for this operation.

"Logged on to workplace" category



The below-mentioned fields pertaining to this category are only filled out, provided that the option "Show the operations logged on for each workplace" has been set.

Workplace

Workplace to which the operation is currently being logged on.

Short name

Short name of the workplace

Designation

Designation of the workplace

Group

Group of the workplace

Cost center

Cost center of the workplace

Toolbar

In general, the parameters for calling the function or target application are taken over from the table. For this reason, an entry should always be selected before calling an application.



Order information

This button opens the application [order information](#).



Order overview

This button opens the application [order overview](#).

7 Finished Operations

Summary

Menu	Order Management --> Production Reports --> Finished Operations
Transaction code	fop
Function authorization	fop

Utilization

The "finished operations" application provides the shift leader, supervisor or foreman with a clearly categorized selection of finished operations.

Integration

The application shows the operations, which were selected in the selection panel.

All operations assigned to the production flags "E" (finished), "A" (archived) and "D" (logically deleted) are displayed.

Selection criteria

The application provides the following selection criteria:

Workplace from... to ...

This selection criterion refers to the workplace planned for the operation. All operations assigned to the selected workplaces are displayed. Wildcards can be used.

Group from ... to ...

This selection criterion refers to the group that is planned for the operation. All operations assigned to the selected group are displayed. Wildcards can be used.

Planned for

This option allows for the displayed operations to be restricted to operations that have initially only been planned for a workplace or a group.

Order

This selection criterion refers to the order number. Operations assigned to the specified order number are displayed. Wildcards can be used.

Article

This selection criterion refers to the article in the operation. All operations assigned to the specified article are displayed. Wildcards can be used.

Article designation

This selection criterion refers to the article in the operation. All operations assigned to the specified article designation are displayed. Wildcards can be used.

Operation status

Current operation status

Predecessor status

Status of the predecessor operation.

Control

Current production flag of the operation

Category

This selection criterion refers to the category of the operation's order type. Only operations the order types of which correspond to the specified category are displayed.

Order type

This selection criterion refers to the order type of the operation. Only operations assigned to the selected order type are displayed.

Processing code

This selection criterion refers to the processing code of the operation. Only operations assigned to the selected processing code are displayed. Wildcards can be used.

Priority

This selection criterion refers to the priority of the operation. Only operations assigned to the selected priority are displayed.

OP

This selection criterion refers to the operation number. Operations assigned to the entered operation number are displayed. Wildcards can be used.

Tool

This selection criterion refers to the tool in the operation. All operations assigned to the specified tool are displayed. Wildcards can be used.

Show split OPs

If this checkbox is checked split operations are displayed additionally.

Customer designation

This selection criterion refers to the customer designation defined in the operation. All orders including the selected customer designation are displayed.

Cost center

This selection criterion refers to the cost center of the workplace defined in the operation. All operations assigned to the selected cost center are displayed. Wildcards can be used.

Planned start between ... and

This selection criterion refers to the planned start date defined in the operation. Only operations planned on or between the selected planned start dates are displayed.

Scheduled start between ... and

This selection criterion refers to the scheduled start date of the operation. Only operations planned on or between the selected scheduled start dates are displayed.

Earliest start between ... and

This selection criterion refers to the earliest start date of the operation. Only are operations displayed, which are planned on or between the selected earliest start dates.

Latest end between ... and

This selection criterion refers to the latest end date of the operation. Only are operations displayed, which are planned on or between the selected latest end dates.

Actual start between ... and

This selection criterion refers to the start date of the order. Only are operations displayed the order start of which is planned on or between the selected dates.

Actual end between ... and

This selection criterion refers to the end date of the order. Only are operations displayed the order end of which is planned on or between the selected dates.

Basic start date between ... and

This selection criterion refers to the basic start date of the order. Only are operations displayed whose basic start time of the order is planned on or between the selected dates.

Basic end date between ... and

This selection criterion refers to the basic end date of the order. Only are operations displayed whose basic end time of the order is planned on or between the selected dates.

Sales order

This selection criterion refers to the sales order defined at the order header. All operations assigned to the selected sales order are displayed.

Order index from ... to

This selection criterion refers to the order index defined at the order header. All operations whose order index at the order header matches the selected order index are displayed

Planned order

This selection criterion refers to the planned order number from SAP that is defined in the order header.

Project number

This selection criterion refers to the project number in the operation. All orders that are assigned to the selected project number are displayed.

Order group

This selection criterion refers to the order group at the order header. All operations are shown whose order group at the order header is assigned to the selected order group. Wildcards can be used.

MRP controller

This selection criterion refers to the MRP controller defined at the order header. All operations whose MRP controller at the order header corresponds to the specified MRP controller are displayed. Wildcards can be used.

MOP

Restricts the merged operations that are displayed. Wildcards can be used.

MOP display

Defines the [merged operations](#) to be used. At first merged operations and individual operations are displayed. Operations summarized in an MOP are not displayed.



In this application, the responsibility area of the planned operations is checked against the workplace which they are planned for.

Detail application: Finished Operations

The table shows the following fields

"Status" category

Status

The bitmap ("LED") defined in the status configuration is displayed as status.

The number of operations is displayed in this column within the totals line.

Status text

The status text results from the current status of the operation.

Status since

Informs about the time since when the corresponding status is set

Predecessor status

Status of the predecessor operation. Here, it can be recognized whether the predecessor operation has already been started and thus material, which will be further processed in the current operation, has already been processed or produced.

Secondary status

Displays the currently set secondary status.



Secondary statuses are configured and used optionally while customizing HYDRA.

If an X is displayed here it is an operation of an order that has been completed technically in SAP.

"Order" category

Displays specific data for [operations](#) and [orders](#).

Relevant fields are:

Order type

Displays the order type as text and symbol. The order types within the HYDRA standard are described in the glossary. Further order types can be defined while customizing HYDRA.

Order

Number of the respective order

Sequence

Order sequence (only relevant if sequences are used)

OP

Number of the respective operation

Split

Split number provided that it is a split operation.

Current number of splits

Number of splits of an operation provided that this one has been split.

Note

Short text of the first note of an operation .

Printed

This column shows whether the corresponding time ticket has already been printed.

"Specifications for production" category

Specifications for production with respect to the machine, tool, DNC, material for the [operation](#).

"Target times" category

Default target time specifications for the [operation](#)

Additionally calculated fields:**Total setup time**

Sum total of setup time, additional setup time and dismantling/teardown time.

Target execution time

Sum total of the total setup time and processing time.

"Processing" category

Default target specifications for the production of the [operation](#)

"Primary quantity/secondary quantity/tertiary quantity" category**Target quantity**

Quantity specification for the [operation](#).

Yield

The yield entered at the terminal or console/MOC is displayed in the yield column.

Scrap

The scrap entered at the terminal or console/MOC is displayed in the "scrap" column.

Rework

Quantity that has to be reworked.

Problem quantity

The problem quantity is another quantity account.

Unit

Quantity unit of the displayed values.



The quantities listed here are displayed as base, primary, secondary and tertiary quantity. In the majority of cases it is reasonable to have only one of these quantity types displayed. The primary quantity is the quantity type in which the entry is made at the terminal.

"OP dates" category

Date specifications for the [operation](#)

"Key figures" category

The formulas used for calculating the values are described [here](#).

"Lock" category**Lock**

Flag indicating whether the operation has been locked. A locked operation cannot be posted anymore and neither appears in the sequencing list at the terminal.

Locked by and on

Indicates when and by whom the operation was locked the last time.

Unlocked by and on

Indicates when and by whom the operation was unlocked the last time.

Toolbar

In general, the parameters for calling the function or target application are taken over from the table. For this reason, an entry should always be selected before calling an application.

**Order information**

This button opens the application [order information](#).

**Order overview**

This button opens the application [order overview](#).

8 Operations

Summary

Menu	Production control --> Production overview --> Operations
Transaction code	op
Function authorization	aop

Utilization

The "operations" application is an important dialog for shift supervisors, assistant foremen or foremen. This application shows operations grouped by their processing status in a clearly structured manner.



Please note in this context that registered operations (status "running") are only shown by their workplace to which they are logged on, provided that the option "Show the operations logged on for each workplace" has been set.

Integration

This application provides the user with a clearly categorized selection of operations. The configurable MOC user interface enables the user to view the information at large, in detail or clearly structured at a glance.

Please note: The progress of an order, i.e. the total of all operations of an order, can be determined using the [order overview](#) application.

Selection criteria

The following selection criteria are available in the application:

Workplace from... to ...

This selection criterion refers to the workplace planned for the operation. All operations assigned to the selected workplaces are displayed. It is possible to use wildcards.



For operations that are logged on, this is not the workplace which the operation is currently logged on to but the workplace which the operation is planned for.

Group from ... to ...

This selection criterion refers to the group that is planned for the operation. All operations assigned to the selected group are displayed. It is possible to use wildcards.



For operations that are logged on, this is not the group of the workplace which the operation is currently logged on to but the group which the operation is planned for.

Planned for

This option allows for the displayed operations to be restricted to operations that have initially been planned for a workplace or a group.

Order

This selection criterion refers to the order number. Operations assigned to the specified order number are displayed. It is possible to use wildcards.

Article

This selection criterion refers to the article in the operation. All operations assigned to the specified article are displayed. It is possible to use wildcards.

Article designation

This selection criterion refers to the article in the operation. All operations assigned to the specified article designation are displayed. Wildcards can be used.

Operation status

Current operation status

Predecessor status

Status of the predecessor operation.

Control

Current production flag of the operation

Category

This selection criterion refers to the category of the operation's order type. Only operations the order types of which correspond to the specified category are displayed.

Order type

This selection criterion refers to the order type of the operation. Only operations assigned to the selected order type are displayed.

Processing code

This selection criterion refers to the processing code of the operation. Only operations assigned to the selected processing code are displayed. It is possible to use wildcards.

Priority

This selection criterion refers to the priority of the operation. Only operations assigned to the selected priority are displayed.

OP

This selection criterion refers to the operation number. Operations assigned to the entered operation number are displayed. It is possible to use wildcards.

Tool

This selection criterion refers to the tool in the operation. All operations assigned to the specified tool are displayed. It is possible to use wildcards.

Show split OPs

If this checkbox is checked split operations are displayed additionally.

Customer name

This selection criterion refers to the customer designation defined in the operation. All orders including the selected customer designation are displayed.

Cost center

This selection criterion refers to the cost center of the workplace defined in the operation. All operations assigned to the selected cost center are displayed. It is possible to use wildcards.

Planned start between ... and

This selection criterion refers to the planned start date defined in the operation. Only operations which are planned on or between the selected planned start dates are displayed.

Scheduled start between ... and

This selection criterion refers to the scheduled start date of the operation. Only operations planned on or between the selected scheduled start dates are displayed.

Earliest start between ... and

This selection criterion refers to the earliest start date of the operation. Only are operations displayed, which are planned on or between the selected earliest start dates.

Latest end between ... and

This selection criterion refers to the latest end date of the operation. Only are operations displayed, which are planned on or between the selected, latest end dates.

Actual start between ... and

This selection criterion refers to the start date of the order. Only are operations displayed the order start of which is planned on or between the selected dates.

Actual end between ... and

This selection criterion refers to the end date of the order. Only are operations displayed the order end of which is planned on or between the selected dates.

Basic start date between ... and

This selection criterion refers to the basic start date of the order. Only are operations displayed whose basic start time of the order is planned on or between the selected dates.

Basic end date between ... and

This selection criterion refers to the basic end date of the order. Only are operations displayed whose basic end time of the order is planned on or between the selected dates.

Sales order

This selection criterion refers to the sales order defined at the order header. All operations assigned to the selected sales order are displayed.

Order index from ... to

This selection criterion refers to the order index defined at the order header. All operations are displayed whose order index at the order header matches the selected order index.

Planned order

This selection criterion refers to the planned order number from SAP that is defined in the order header.

Project number

This selection criterion refers to the project number in the operation. All orders that are assigned to the selected project number are displayed.

Order group

This selection criterion refers to the order group at the order header. All operations are displayed whose order group at the order header is assigned to the selected order group. It is possible to use wildcards.

MRP controller

This selection criterion refers to the MRP controller defined for the order header. All operations are displayed whose MRP controller of the order header corresponds to the entered MRP controller. It is possible to use wildcards.

MOP

Restricts the merged operations that are displayed. Wildcards can be used.

Show MOP

Defines the [merged operations](#) to be displayed. Initially, merged operations and individual operations are displayed; operations combined into an MOP are not displayed.

Show the operations logged on for each workplace

If this option is checked, an operation that is currently being logged on to multiple workplaces will be shown as many times as it is actually logged on.

The category "logged on to workplace" (hidden by default) shows information about the workplace which the operation is logged on to.



In this application, the responsibility area of the planned operations is checked against the workplace which they are planned for.

"Operations" detail application

The table shows the below-mentioned fields, among other things. Information on further fields can be found in the documents dealing with

- [data about the backlog of orders](#)
- [current data and key figures on the operation](#)

"Status" category

Status

The bitmap ("LED") defined in the status configuration is displayed as status.

This column shows the number of operations within the totals line.

Status text

The status text results from the current status of the operation.

Status since

Informs about the time since when the corresponding status has been set

Predecessor status

Status of the predecessor operation. Here, it can be recognized whether the predecessor operation has already been started and thus material, which will be further processed in the current operation, has already been processed or produced.

Secondary status

Displays the currently set secondary status.



Secondary statuses can be configured and used optionally as a part of HYDRA customizing.

"Order" category

Displays specific data for [operations](#) and [orders](#).

Relevant fields are:

Order type

Displays the order type as text and symbol.

The standard order types are described in the glossary. Further order types can be defined while customizing HYDRA.

Category

The category summarizes similar order types. The following categories are available in HYDRA:

FA = production order

PJ = project order

PM = maintenance order

KP = capacity order

GK = overhead cost order

Order

Number of the corresponding order

Sequence

Order sequence (only relevant if sequences are used)

OP

Number of the respective operation

Split

Split number provided that it is a split operation.

Current number of splits

Number of splits of an operation, provided that this one has been split.

Note

Short text of the first note of an operation.

Printed

This column shows whether the corresponding time ticket has already been printed.

"Specifications for production" category

Relevant fields are:

Planned workplace

Workplace on which the operation is planned.

Planned for

The "planned" column shows whether the operation is planned

G at a group (pool of groups) or

M at a workplace (pool of workplaces, machines).

Group

Group for which the operation is planned.

"OP dates" category

Date specifications for the [operation](#)

"Remaining run time" category**Remaining run time**

Still remaining production time. This is a calculatory value that is calculated by a formula based on different parameters. The formula is defined in the operation.

Remaining run time

Still remaining production time. This is a calculatory value that is calculated by a formula based on different parameters. The formula is defined in the operation.

"Target times" category

Default target time specifications for the [operation](#)

Additionally calculated fields:**Total setup time**

Sum total of setup time, additional setup time and dismantling/teardown time.

Target execution time

Sum total of the total setup time and processing time. **"Primary quantity/secondary quantity/tertiary quantity" category**

Target quantity

Quantity specifications for the [operation](#).

Yield

Yield posted onto the operation.

Scrap

Scrap posted onto the operation.

Rework

Quantity that has to be reworked and that is posted onto the operation.

Problem quantity

Problem quantity posted onto the operation.

Unit

Quantity unit of the displayed values.



The quantities listed here are displayed as base, primary, secondary and tertiary quantity. In the majority of cases, it is reasonable to have only one of these quantity types displayed. The primary quantity is the quantity type in which values are entered at the terminal.

"Postings" category

First logon

Date and time when the operation was logged on for the first time.

Last logoff

Date and time when the operation was logged off the last time. This kind of information can be interesting if an operation was reactivated.

Date of last posting

Date of the last posting made for this operation.

Time of last posting

Time of the last posting made for this operation.

"Logged on to workplace" category



The below-mentioned fields pertaining to this category are only filled out, provided that the option "Show the operations logged on for each workplace" has been set.

Workplace

Workplace to which the operation is currently being logged on.

Short name

Short name of the workplace

Designation

Designation of the workplace

Group

Group of the workplace

Cost center

Cost center of the workplace

Toolbar

In general, the parameters for calling the function or target application are taken over from the table of the "operations" detail application. For this reason, an operation should always be selected before calling an application.



Generate merged operation

[Generate merged operation](#) function Selected operations are consolidated to a single merged operation. The consolidated operation is representatively logged on and displayed at the terminal

**Cancel merged operation**

[Cancel merged operation](#) function

The merged operation is broken up into separate operations.

**Order information**

This button opens the application [order information](#).

**Order overview**

This button opens the application [order overview](#).

**Schedule controlling: save baseline plan**

Function [Schedule controlling: save baseline plan](#)